

Advaya Verma

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EDUCATION

Vellore Institute of Technology, Andhra Pradesh — B. Tech, Computer Science and Engineering | 2023 – 2027

CGPA: 8.1

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Machine Learning, Artificial Intelligence, Software Engineering & UML, Theory of Computation, Compiler Design, Design & Analysis of Algorithms

EXPERIENCE

AI Engineering Intern — IDM Valley | Apr 2026 – Jun 2026

- Applied semantic search and information-retrieval concepts from prior project work to develop and experiment with Retrieval-Augmented Generation (RAG) workflows for LLM-based applications.
- Worked with text embeddings, document retrieval, and vector databases to build retrieval pipelines for supplying relevant context to language models.
- Experimented with prompt engineering and context integration to improve the quality and relevance of LLM-generated responses.
- Used Python for data preprocessing, retrieval workflow development, experimentation, and integration of AI components with application backends.
- Created technical documentation covering RAG workflows, implementation approaches, configuration, and usage guidelines to support development and knowledge sharing.

PROJECTS

Production-Style RAG Knowledge Assistant | Python, FastAPI, FAISS, BM25, Sentence Transformers, RAG, Gemini, React, Docker | [GitHub](#)

- Built a production-style RAG pipeline with page-aware document ingestion, configurable chunking, 384-dimensional Sentence Transformer embeddings, FAISS dense retrieval, and BM25 sparse retrieval for semantic and keyword search.
- Implemented two-stage hybrid retrieval using RRF/weighted fusion and Cross-Encoder reranking, with grounded LLM responses, structured source citations, and automated evaluation using Recall@K, Precision@K, MRR, NDCG, MAP, and latency benchmarks.
- Developed a two-tier semantic caching system with exact hash and FAISS vector matching, LRU/TTL-based lifecycle management, and cache invalidation; integrated a React frontend, Dockerized deployment, REST APIs, and end-to-end performance benchmarking.

Semantic Search System | Python, FastAPI, FAISS, Docker, NLP, Scikit-learn | [GitHub](#)

- Built a semantic search pipeline using Sentence Transformers (MiniLM-L6-v2) and FAISS, achieving sub-100ms nearest-neighbor retrieval over 18,000+ documents from the 20 Newsgroups dataset.
- Implemented Gaussian Mixture Model (GMM)-based fuzzy clustering to assign probabilistic multi-topic memberships to documents, improving retrieval relevance for ambiguous queries.
- Designed a semantic cache layer to detect similar queries and reuse results, reducing redundant API calls and cutting average response time by ~40%.
- Exposed REST API endpoints via FastAPI and containerized the full application using Docker, enabling one-command portable deployment.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, R, JavaScript

Web & Backend: HTML, CSS, React, FastAPI, REST APIs

ML / AI: Scikit-learn, NumPy, Pandas, Sentence Transformers, FAISS, NLP, RAG, LLMs, Generative AI

AI Concepts: Embeddings, Semantic Search, Vector Databases, Clustering

Tools & Platforms: Docker, Git, GitHub, MySQL, Oracle Cloud Infrastructure

Software Engineering: Agile/Scrum, Software Development, UML

CERTIFICATIONS

Hashgraph Developer Course — The Hashgraph Association | Mar 2026

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle University | Oct 2025